Quick Response Code Validation and Phishing Detection Tool

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***Abstract* —** A Quick Response (QR) Code is a type of barcode that can be read by the digital devices and which stores the information in a square-shaped. The QR Code readers can extract data from the patterns which are presented in the QR Code matrix. A QR Code can be acting as an attack vector that can harm indirectly. In such case a QR Code can carry malicious or phishing URLs and redirect users to a site which is well conceived by the attacker and pretends to be an authorized one. Once the QR Code is decoded the commands are triggered and executed, causing damage to information, operating system and other possible sequence the attacker expects to gain. In this paper, a new model for QR Code authentication and phishing detection has been presented. The proposed model will be able to detect the phishing and malicious URLs in the process of the QR Code validation as well as to prevent the user from validating it. The development of this application will help to prevent users from being tricked by the harmful QR Codes.

***Keywords* —** *Quick Respond Code, Malicious URLs, QR Code Phishing Detector, Pharming and File Transfer Protocol.*

I. INTRODUCTION

In 2020, the use of the internet in life is growing rapidly. This is due to the pandemic Covid 19. Many companies have changed the mode of work to work from home, the students of higher learning institutes and school students have also switched to online learning. The effects of movement control instructions from the government, banking activities and the purchase of goods or services also have shifted to online form. [1]. Due to the increasing of internet usage in our daily lives, it also opens up opportunities for cyber criminals to make a profit. This is a kind of threat to the internet users nowadays. There are various forms of internet-related threats, such as malicious programs, some of which use host programs such as trap doors, Trojan horses, logic bombs and some separate ones, such as viruses, worms and zombies. Social Engineering techniques along with phishing is one of the most recurrent cyber threat, that people can easily fall to it[2]. Among of the mentioned threats, phishing attacks have become dramatically more common in the world in recent years. Phishing is the way to attract people to visit and convince fraudulent websites to enter their personal details on the same websites. [3-5].

Phishers normally create clone web pages to spoof users. Phishing is considered as a semantic attack and commonly interpreted as online identity theft. Phishing attacks based on QR (Quick Response) codes is the most recent oxygenating factor for the attackers. For the common user, they usually do not know the content behind a QR code until they access the web or display the data behind it. This encourages users to inadvertently download malware or applications that contain viruses. Other than that, the QR code itself is not equipped with thorough protection which means there are vulnerabilities exploited for attacker purposes. [6-9]. Therefore, the main objective of this work is to come out with a model that can detects a phishing QR code by enhancing the security of the QR code scanner and implement an algorithm that can blocks malicious links or URL.

II. LITERATURE REVIEW

The term phishing initially introduced in 1990s [10]. It was created from the word 'fishing' where the attacker also known as phisher directs user to visit fake website. The common method used by the phisher is sending fake emails containing a URL pointing to a fake website. The main goal of phishing is to obtain internet user personal information such as user name, password, national security ID or bank account number [11] where later on they will use this information to conduct transactions on behalf of the victim [12]. Other than email, phishing is also spread through VOIP, SMS, Instant messaging, and social networking sites and even through multiplayer games. Phishing can be categorized into several type such as clone phishing, spear phishing, phone phishing, SMS phishing, DNS-based phishing and man in the middle attack phishing. A clone phishing is a form of phishing attack in which a phisher attempts to clone a website regularly visited by the victim [13]. The clone website generally asks for login credentials, to imitate the real websites. Spear phishing targets a small community, so spear phishers target groups of people with something in common instead of casting out thousands of e-mails at random.

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[14]. Phone phishing refers to a message claiming to be from a bank asking users to dial a phone number about bank account problems. A type of phone phishing is SMS phishing. The victims receive an SMS telling that he/she has subscribed to a service successfully. The victim can visit the website and need to provide confidential details if he/she wants to cancel the service. The next phishing form is known as DNS-based phishing (pharming). This attack uses a fake website to redirect victims to another fake website. Pharming interferes with domain name resolution to an IP address such that the actual website's domain name is mapped to the rogue websites' IP address. A man in the middle attack also refers to an attack in which the electronic messages given between the sender and the recipient are secretly intercepted by an intruder. Then, during message transmission, they record, insert and change messages. A middle-attack man uses Trojan horses to intercept personal data. [15].

Studies have shown that QR codes for many security threats can be used as an attack vector. A proof of concept phishing attack using QR code was performed by Keisberg et al. They also illustrated the dangers of potential attacks using manipulated QR codes in their research and highlighted that proper input sanitization must be carried out before processing the contained data. [16]. This threat occurs because the website address (URL) can be encoded in a QR code. Some websites use QR codes with their login page URLs encoded in them. In addition, QR codes are also often used in advertisements to direct the target audience to special offers or additional information about a particular product. Thus, an attacker can replace the real QR code with a fake QR code causing the user to be redirected to an unintentional page that may contain malware that will be automatically downloaded [17]. Amin et. al in their study of malicious QR codes state that they have found about 150 malicious QR codes that were designed to redirect victims to phishing sites or redirect users to either exploit or intermediate sites. Spoofed versions of password-protected websites, fake versions Google Play app market, malware distributed via direct download links were the common threats [16].

One of the general issues involving the QR code as they are turning into a well-known method for encoding data is the SQL infusion. Consider a situation where a scanner is associated with a database and QR code data are used to execute an inquiry on the backend database. In such a situation, if the QR code contains an inquiry, for example, "1' OR '1'='1" then the reader will execute the query without checking whether it comes from an authenticated source or not, which can lead to the show of information to a possible hacker intended for an authorized user. Browser based exploits and XSS attacks are another concern. To perform program-based assaults and cross site scripting assaults, a QR code could be used. We know that there could be an encoded URL in a QR code. Presently, presume that the encoded URL contains an alert message containing an effort dependent on a program. In such an example, the effort is encoded inside the alarm when a client gets to that URL. Consider a scenario where the input used is a command line parameter from a QR code. In such a scenario, by modifying the QR code and thus executing arbitrary commands on the machine, an attacker may easily exploit this. This allows an attacker to install rootkits, spyware, start a Denial-of-Service (DoS) attack, or attach a shell to a remote machine and access it from there. [18-20].

III. PROPOSED METHOD

The proposed work concentrated on identifying phishing URLs and valid URLs code. The method used is shown in Fig. 1. The process start with reading the QR code. If the application detected the QR code image, it will start identify and examine whether it is a legitimate or phished QR code. If it is legitimate URL then it will redirect the user to the site. Otherwise the application will display warning message to the user. The process of detection is based on the work done by [21].

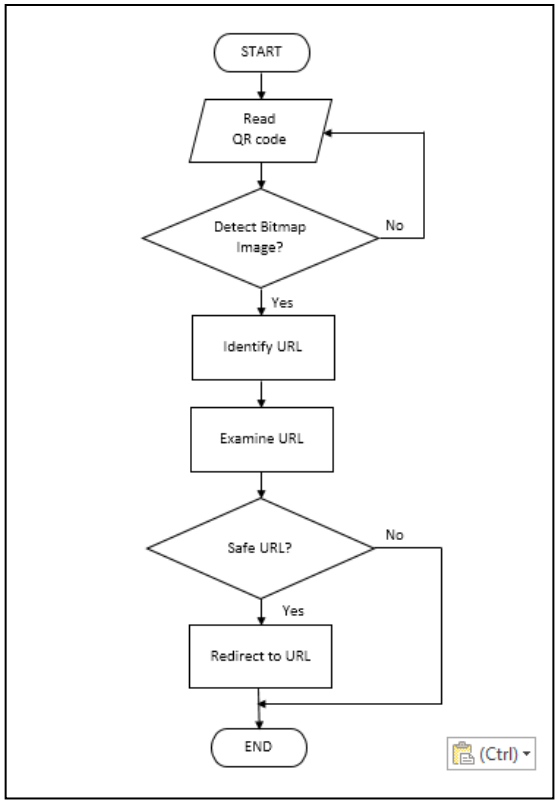

Fig 1: QR Code Phishing Detection Process

A. *Phishing Detection Rules*

One of the rules used to identify the genuine and phished URL is through the length of the host URL. A

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URL is a formatted string used for defining a network resource on the internet by internet users. There are three components to the URL string, such as network protocol, host name, and route. For phishing and valid URLs, domain name size is examined. In phishing URLs, the standard domain name size is found to be greater than 25 characters and the average valid domain name size is found to 20 characters' length.

The second rule used to check the valid and phished URL is through the number of slashes in URL. The phishers attempt to deceive web clients by impersonating the dubious URL look authentic. One such procedure utilized in defrauding is the expansion of cuts in URL. The present investigation in this manner, considers the number of slashes in URLs as an element of distinguishing proof of phishing and look at the number of slashes in genuine and phishing URLs

The third rule used is through the dots in the host name of the URL. This rule check on the number of dots in the host name of the URL. Phishing URL for the most part have numerous dots to cause clients to accept that they are real page. Results have indicated that the suspicious URL has multiple dots and the quantity of dots in an authentic URL is normally three.

The forth rule used is through the number of terms in the host name of the URL. Domain names are utilized to identify unique identity. The host name part is separated from the URL. The terms in the host name segment are tokenized. On the off chance that the quantity of terms in the host name segment of the URL is halfway, at that point the URL is genuine generally phishing. The identity of the authentic site is gotten from the host name of the URL. It was discovered that 77.75% of phished URLs contain special characters. It also was discovered that 9.4% of phished URLs contained IP address.

The fifth rule used to identify between legitimate and phished URL is through examining the Unicode in URL. Unicode gives an exceptional number to each character. It is investigated that the greater part of the phishing URLs are coded with Unicode characters. The nearness of Unicode in the host name of the URL demonstrates that the URL is a phished URL. It was discovered that 65.16% of phished URLs contained Unicode characters. The URL is separated into two components, host component and a path component. The URL utilizes transport layer security to decide if the URL is ensured. The nearness of HTTPs convention is required when fragile data is moved across organize. Studies shows that 99.0% of URLs were found without transport layer security [21].

Another feature that was taken into count was the PhishTank based feature. Every month, PhishTank publishes statistical reports on phishing websites. The purpose is to use historical data on the top IPs and domains that host websites for phishing. If the URL host is part of the IP or top domain that hosts the phishing site and many heuristics related to phishing are available, then it is likely to increase the likelihood of classifying the URL as phishing.

The proposed method analyses the lexical based feature such as the presence of suspicious characters encoding the URL, hexadecimal characters and malicious IP addresses to hide them. Various hosts based features such as the age of the domain, page rank and domain confidence level are also analyzed to avoid the end users from falling prey to the phishing attacks. The properties of Lexical features used to detect the attacks include the length of URL, the length of the hostname and the number of dots in the URL.

B. URL Identification

HTTPs stands for Secure Hypertext Transfer Protocol. This protocol is used to secure the communication through a computer network using the Transport Layer Security (TLS) or formerly its predecessor, Secure Sockets Layer (SSL). The protocol is therefore also often referred to as HTTP over TLS or HTTP over SSL. The Uniform Resource Identifier (URI) scheme HTTPs has identical usage syntax to the HTTP scheme. However, HTTPs signals the browser to use an added encryption layer of SSL / TLS to protect the traffic. HTTPs creates a secure channel over an insecure network. This ensures reasonable protection from eavesdroppers and man in the middle attacks.

The File Transfer Protocol (FTP) is a standard network protocol used to transfer computer files between a client and server on a computer network. FTP is built on a client-server model architecture using separate control and data connections between the client and the server. FTP users may authenticate themselves with a clear-text sign in protocol. Other than that, FTPS is a commonly used protocol that adds support for the Transport Layer Security. FTPS is a secure file transfer subsystem for the Secure Shell (SSH) protocol with which it is not compatible.

The Gopher protocol is a TCP/IP application layer protocol created for distributing, searching and retrieving documents over the internet. The Gopher protocol was strongly oriented towards a menu – document design and presented an alternative to the World Wide Web in its early stages. The Gopher protocol uses the TCP port 70.

C. Application Development

The QR Code Phishing Detection tool has been developed using Android Studio and Java NetBeans IDE. An Android based device will be needed to implement the application and must be 4.0.3 and

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above. Fig 2 shows the main interface of the application. This interface contains scan button that will redirects the user to the scanning QR code interface as shown in Fig 3.

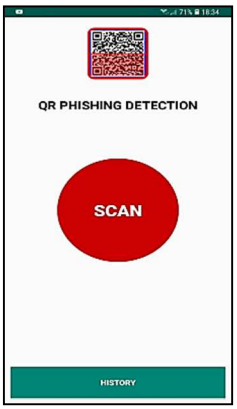

Fig 2: Main interface of QR Code Phishing Detection Tool

Fig 3: Scanner Interface

Upon scanning if the link is a safe link the application will display a message to user saying it's a safe link with option for the user to choose whether to proceed to open the link or to cancel the scanning as shown in Fig 4. Upon clicking the open button, the application will redirect the user to the scanned web page while clicking the cancel option, the application will redirect the user back to the earlier page to scan a new QR code. If the application manages to detect the phishing link, the application will immediately alert the user that the scanned QR code is dangerous and it might be a phishing link followed by the reasons for the link is blocked by the application (Refer to Fig 5).

The reason can be the scanned links domain is suspicious, the URL string patterns doesn't look safe and is abnormal, the port number of the website doesn't match the existing ports, the IP address string is not matching with the existing patterns. Upon this, the application will allow the user to validate the scanned link, other than that an option button will be created to cancel and go back to scan a new QR code.

Fig 4: Valid QR Result Interface

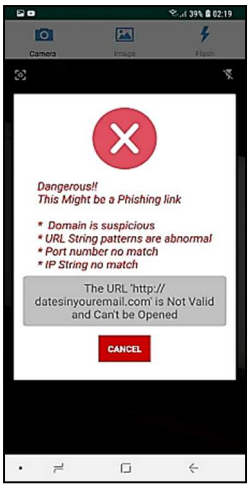

Fig 5: Invalid QR Result Interface

The test plan has been conducted to fulfil the requirements of the QR Code Phishing Detection Tool. Table 1 shows the features that has been tested and also the results.

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Table 1: Test plan of the QR Phishing Detection Model

Features	Results	Status
User can scan QR code	The application was able to scan the QR codes	Success
Identify the status of URL	The application was able to detect the safe links and also the malicious links	Success
Able to redirect to URL	The application was able to redirect users to the safe URL once scan is done.	Success
Open gallery in device	The application was able to open the gallery in the device and scan the saved QR codes.	Success
Open flashlight	The flashlight was able to be used to scan even brighter	Success

IV. CONCLUSION

In conclusion, phishing is one of the online platforms to steal information from internet users not only to steal personal information of users but also information related to online transactions especially online banking passwords and credit card information. Phishing can now be implemented through the use of QR Codes. It is because QR code is versatile and convenience as well as much more appealing. However, insufficient emphasis on QR Code security causes problems such as sensitive data leakage and poor protection against malicious attacks. The development of the QR Code phishing detection tool will enhance QR Code security to a higher stage and forbid QR Code abuse by attacker. This project is mainly for the users who scans the QR codes around them in their daily activities, such as students or employees and this tool will make sure their information will not be leaked or breached when scanning a QR code. The advantage of this detection model is it detects the abnormal and malicious links that are embedding in the QR codes to be used as a vector attack. This QR code phishing detection tool was developed and tested before the tool was presented. In order to make sure the scanning is done perfectly and the malicious and phishing page is detected, the protocols that has been used is HTTP, HTTPs, FTP, SFTP, RTSP, Telnet, NNTP, Gopher. The algorithms that was used is Patterns.Web_URL.matcher and URLUtil.isValidUrl. With this the application has also been developed to examine the domain, IP address, string patterns and also the host name of the scanned link before producing the results. This implementation has helped in detecting and also securing the application for a safer scanning to be done. This application will enable user to experience a better QR Code scanning and protect them from malicious URLs. An attacker can use QR Codes to do a fraud. When user scan the QR code, they may get redirected to a fake web site unknowingly. The development of this Phishing detection tool will ensure user does a secure scanning and this will enhance the security of QR codes.

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